Pratik Musale

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Education

University of Pittsburgh *PhD in Computer Science* Aug 2019-Present
- Advisor: Dr. Adam Lee
- Thesis: Privacy in Smart Spaces: Exploring the Interplay Between User Preferences, Awareness Requirements, and Secure Technologies.

Stony Brook University *MS in Computer Science* June 2018
- Advisor: Dr. Bong Jun Choi
- Thesis: Lightweight Authentication for IoT Ecosystem

Savitribai Phule Pune University, India *BE in Electronics and Telecommunication* May 2015
- Advisor: Dr. Pranoti Mane
- Project: Accident Prevention System in Cars

Awards

Ph.D Fellowship, University of Pittsburgh Aug 2019-May 2020

Publication

[1] **Musale, Pratik** and Adam Lee. "Trust TEE?: Exploring the Impact of Trusted Execution Environments on Smart Home Privacy Norms". In: *Proceedings on Privacy Enhancing Technologies* (2023).

[2] Duin Baek, **Musale, Pratik**, and Jihoon Ryoo. "Walk to Show Your Identity: Gait-based Seamless User Authentication Framework Using Deep Neural Network". In: *The 5th ACM Workshop on Wearable Systems and Applications*. WearSys '19. Seoul, Republic of Korea: Association for Computing Machinery, 2019, pp. 53–58. ISBN: 9781450367752. DOI: 10.1145/3325424.3329666. URL: https://doi.org/10.1145/3325424.3329666.

[3] **Musale, Pratik**, Duin Baek, Nuwan Werellagama, Simon S. Woo, and Bong Jun Choi. "You Walk, We Authenticate: Lightweight Seamless Authentication Based on Gait in Wearable IoT Systems". In: *IEEE Access* 7 (2019), pp. 37883–37895. DOI: 10.1109/ACCESS.2019.2906663.

[4] **Musale, Pratik**, Duin Baek, and Bong Jun Choi. "Lightweight gait based authentication technique for IoT using subconscious level activities". In: *2018 IEEE 4th World Forum on Internet of Things (WF-IoT)*. 2018, pp. 564–567. DOI: 10.1109/WF-IoT.2018.8355210.

Projects

User Aligned Configurations in Workplace 2024-Present
- Surveyed 435 participants with CI-framed device configurations for four IoT devices across three hypothetical multiuser workplace scenarios; computed five algorithmic defaults from group preferences.
- Identified Community Coherence phenomenon: users gravitate toward Mid-Ranked and Least-Negotiated in comparison to defaults that were Privacy-Centric, Excessive in Sharing, or based on Majority-Voting. This reflected social moderation over personal assertion.
- Found user preferences structured by three phenomena, privacy risk, utility benefits, and space alignment, making workplace configuration conflicts anticipable rather than random.
- Tools: Qualtrics, IBM SPSS, Python

Configurations with IoT Devices in Workplace Meeting Room 2024-Present
- Conducted five focus groups (N=20) examining how individuals and groups configure IoT devices in a workplace meeting room, presenting CI-structured configurations as flashcards across video conferencing and

in-person team meeting activities.

- Found that individually, users configure using identification risk and limited data retention as primary heuristics; in group settings, users negotiate conditional, activity-driven, and asymmetric configurations that existing access control mechanisms cannot represent.
- Identified two security risks, policy collapse and configuration-induced insider exfiltration, and contributed four design implications, including Group CI-Profiles for preserving group-negotiated agreements.
- Tools Used: NVivo

Privacy Awareness in Smart Workplaces 2024-Present

- Surveyed 399 American adults on privacy information prioritization across four workplace spaces and four IoT devices; found data type and sensing purpose consistently prioritized over other CI parameters.
- Designed four interface prototypes (notice, push notification, dashboard, AR app) presenting CI information in a two-layer hierarchy; conducted 15 semi-structured interviews across the four spaces.
- Found users prefer dashboards for collaborative spaces and AR for individual spaces, driven by social norms and space usage patterns.
- Tools: Qualtrics, IBM SPSS, Python, NVivo, JavaScript, Android, Unity (AR)

Impacts of Secure Computing in Smart Homes 2022-2023

- Between-subjects survey of 1,049 American adults (with-TEE vs. without-TEE) on IoT information flows for smart speaker and camera across device manufacturer and subject (user vs. bystander) conditions.
- TEE reduced notification fatigue and collapsed the audio/video distinction in users' mental models; however, TEE did not resolve bystander privacy concerns, showing secure technology alone cannot address multi-user smart space complexity.
- Tool Used: Qualtrics, IBM SPSS, Python

Gait Authentication 2016-2018

- Authenticating users every time for them to interact with IoT can be inconvenient to the users. Additionally, deploying specialty sensors for continuous authentication can be a cumbersome task for both the users and the organizations. We conducted a study with 49 participants to collect gait profiles of users.
- The project focuses on understanding the use of smartphones and smart watches to identify walking patterns. Furthermore, the project utilizes the onboard sensors available within the smartphone and the smartwatch.
- Additionally, the project tries to reduce the features required to identify walking patterns to make it lightweight.
- Tools Used: Android, JavaScript, NoSQL, Python, Machine Learning Packages SciKit and TensorFlow

Undergrad Project - Accident Prevention System in Cars 2013-2015

- Accidents are mainly coming from road rage, drinking or not paying attention cause of the media volume. However, to monitor drivers for their patterns, and use of the cars is till date restricted,
- The project designed a system using ATMEGA 16 to continously monitor driving patterns using an Accelerometer, warning on media volume by using Decible Meter, and faciltating a message with car location coordinates on accident or for any abuse to person in the car.
- Technologies Used: OpenPLC Editor, VHDL, MATLAB, Embedded Programming

Appointments

Department of Computer Science, University of Pittsburgh
Graduate Teaching/Research Assistant June 2020-Present

Department of Computer Science, Stonybrook University
Graduate Teaching/Research Assistant Aug 2016-May 2018

Work Experience

Navigate Consulting - [Security Researcher Intern] Jan 2019-June 2019

- To investigate the feasibility of single sign-on feasibility for the Industry IoT ecosystem.

- Designing prototypes for the deployment of IoT devices in the field for a metal fabrication plant.

Spectra Televentures - [OMCR Engineer] Sep 2015-Jul 2016

- Maintenance of Radio Towers for Nokia Networks' customer Vodafone.
- Installation and commission of new 3G and 4G network.

Teaching Experience

Teaching Assistant – University of Pittsburgh

- CS 1650: *Software Engineering* (Summer 20)
- CS 1566: *Computer Graphics* (Fall 20)
- CS 1550: *Intro to Operating System* (Spring 21, Summer 21)
- CS 449: *Intro to System Software* (Summer 22, Summer 23, Fall 23, Summer 24, Fall 24)
- CS 1632: *Software Quality Assurance* (Fall 22, Spring 23, Spring 24, Spring 25)
- CS 1653: *Applied Cryptography Network Security* (Summer 23)

Teaching Assistant – Stonybrook University

- CS 214: *Data Structures* (Fall 16)
- CS 114: *Introduction to Object-Oriented Programming* (Fall 17, Spring 18)
- MEC 102: *Engineering Computing and Problem Solving* – Dept of Mechanical Engineering (Spring 17, Fall 17)
- CS 102: *Introduction to Web Design and Programming* (Spring 18)

Additional Course Conducted

- Pitt Air Force Association CyberCamp – Teaching Cybersecurity ethics, principles to high school students (Summer 24, Summer 25)